

SYSTEM REPORT (MOBILE MONEY AGENT COMMISSION TRACKING SYSTEM)

1. APP OVERVIEW

- Mobile money agent commission tracking system; this is a web application for mobile money business owners and their agent employees. For business owners to track the progress of their agents and for agents to record their daily transaction commission.
- This system is made for mobile money (TNM Mpamba/ Airtel Money) business owners who employed people to work for them as agents. And the simplifies the management of those agents in a way that the agents record commission made per day, and the business owners are able to follow up
- The system allows the business owner to add newly employed agent into the system as the business grows, edit their information and delete employees account if they are no longer available.

2. DESIGN DECISIONS

- The system has been built with a HTML, CSS, PHP and mysql tech stack; HTML and CSS for frontend tech with the reason being it is a human related language making it easy to understand and structure the web pages, CSS is easy for styling the web pages and positioning of objects on the page. PHP has been used to interact pages and the database, for example, storage of data in the databases, because PHP code runs on the on servers allowing data from the web pages to be stored on the servers, and also it allows features like authentication and data-encryption for security. And mysql for database management.
- CRUD has played a big role in the business owners side for management of every user of the system, by allowing the owner to see all user accounts, create a user account for new employees, editing their when required and delete if the owners pleases to do so. The system contains a users list table, which contains a user's ID, Name, Email and Role, and from there the business who is the admin in this case can perform the action they require. The Create elements has been implemented as the 'Add' button of which it provides a form of which the admin is required to fill the new user's information that is stored into the database. The Read element is the one that shows the information of every user on the table, it retrieves data from the database and shows it in the table. The Update element works on 'edit' button where the admin clicks the edit button in the rows of the user they want to edit their information and they are provided with a form which is filled with the existing information and so they can edit it and submit it to the database. The system retrieves data of user by matching the email in the rows of the table with the email in the database row and fills it in the edit form. Finally, the Delete element, this one also matches the email in the table with the email in database and deletes the whole rows.
- The system also has an authentication feature where as every user logs into the system with their registered email and password. This has been achieved by first providing the

login form which also has a link a sign up form for creating a new account if the user doesn't have one, the sign up form requires the user to enter their name, email, their role in the business and password they are going to be using for logging in which is then stored in **users** database table. When the user clicks the sign up button, the system checks if the email is already registered in the database and if not, the account is created and the user is directed to the login page, but if the email is already existing in the database, the user is notified that the email is registered to another account. Now when the user wants to login they are required to enter their registered email and password, now the system checks if the entered email is registered in the database and if it is the entered password is checked if it corresponds to the one in rows of the entered email, if not the page is refreshed without logging in, and also when the email is checked, the system also checks the role of which of the account is assigned to. If the email is assigned as a 'Manager' the user is redirected to manager page and if it is assigned to 'Agent', the user is redirected to the agent page.

3. CHALLENGES AND SOLUTIONS

- There were many technical challenges faced in the development stages, some of the problems were making database data appear in the tables in the application, data retrieval was being done in a different **php** file which was challenging to make the data appear a table which was in a different file. As tricky as it was the problem was overcome, and we had to write the php data retrieving code inside table itself and print the data in their required rows. Another challenge was updating user data that is already existing most especially when it came to a user pressing the edit button and they should be redirected to a form that is filled with data of the exact user that has been clicked. For this problem, we had to send the email in the clicked row to the editing form and from there used the email to retrieve data from database and pasted it in its required places so that the admin can just edit the data that already have and save it.

4. TESTING & REFLECTION

- For testing, we used manual and indevelopment testing, every module was tested during development to minimise error when testing the whole system. This method was used in order to make sure that every module is performing atleast the idea of what it was built for.
- If it was that we had more time than we had this time, we could have made some improvements like displaying errors and message if required on the same page instead of showing another page just for an error or message and making them appear in required areas where they could have appeared as errors or important messages. Another improvement could have been creating confirmation messages with "AGREE" and "CANCEL" before performing a certain action in forms and CRUD actions.
- There was also a lesson to be learnt by building this mini application, the lesson being the hardest part of building a software is identifying the exact thing that you want to build. Once you can figure out what you want to build, what features you want to put in the software, what you want to put in your software and where exactly do you want

to put it, the development itself becomes easy as long as you are familiar with the language you are using.